Phase 1: Understanding Product & Users

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Step 1

Getting started with the Product Analysis - Flows & Frictions

S. No	Flows / Touchpoints	What works well	Confusion / Friction
1	Signup → login	Social login → Fast entry	• Unclear value proposition
2	Create Notebook → Add Sources	Easy Upload	• Unclear supported formats • Large file could take time
3	Chat with Notebook	Grounded AI responses with citations	• Responses can be too generic for different research goals and often require prompt refinement.
4	Generate Study Content	Instant summaries	• Generic outputs • Limited customization
5	Share → Revisit Notebook	Simple sharing	• Basic collaboration • Difficult multiple notebook organization

Step 2

Company context (Mission, Revenue & Differentiators)

S. No	Field	Context
1	Mission	Help users understand, organize, and interact with their information using AI grounded in their own sources.
2	Revenue streams	Included with select Google AI plans and enterprise offerings; drives adoption of the broader Google AI ecosystem.

3	User value	• AI grounded in user-uploaded sources • Instant summaries, study guides & audio overviews • Supports multiple document formats • Simple, distraction-free interface
4	Business value	Increases engagement with Google's AI ecosystem and Workspace. Encourages users to store and work with their knowledge in Google products, creating opportunities for premium AI subscriptions and enterprise adoption.
5	Key Differentiators	1. ChatGPT: General-purpose AI that relies on conversation and optional uploads. NotebookLM is source-first, grounding every response in user documents with citations. 2. Perplexity: Excels at web search and real-time information. NotebookLM focuses on understanding a user's private knowledge base rather than the public web. 3. Notion: Built around workspace productivity and document editing. NotebookLM is designed specifically for research, learning, and synthesizing information from multiple sources.

Step 3

Feature Reverse-Engineering

S. No	Feature	Problem it solves	How it works	Why it works
1	Grounded AI Responses	*"I don't trust AI because it hallucinates or gives incorrect information."*	AI answers questions using only the user's uploaded sources and provides citations.	• For Users: More trustworthy answers with verifiable references. • For Business: Builds user trust and

				differentiates NotebookLM from general-purpose AI assistants.
2	Audio Overview	*"I don't have time to read long documents."*	Converts uploaded content into an AI-generated podcast-style conversation summarizing key ideas.	For Users: Learn while commuting or multitasking. For Business: Increases engagement and showcases Google's multimodal AI capabilities.
3	One-Click Study Materials	*"Creating summaries, FAQs, or study guides manually is time-consuming."*	Automatically generates summaries, FAQs, timelines, briefing docs, and study guides from uploaded sources.	For Users: Saves time and improves learning efficiency. For Business: Encourages repeated usage across students, researchers, and professionals.

Step 4

User segmentation

S. No	Primary segment	Sub-segments	Key needs / Use cases	User Problems
1	Professional users	Researchers / Analysts	Analyze multiple documents, synthesize information, generate reports	Difficulty organizing many sources. AI misses cross-document insights. Limited collaboration.
		Product Managers / Consultants	Research, PRDs, meeting preparation, competitive analysis	Responses often need to be reformatted into reports or presentations.

2	Education users	Teachers / Instructors	Create learning material, explain concepts, prepare lectures	Limited customization of generated content. Basic sharing and collaboration.
		Students	Study notes, exam preparation, summaries	AI explanations aren't always tailored to their learning style or study objective.

Step 5

Problem Hypothesis

Priority	Problem Hypothesis	Why I picked it	Gut confidence
1	If NotebookLM generates responses that better align with users' research objectives, users will spend less time refining prompts, complete research tasks faster, and rely on NotebookLM more frequently.	Users frequently receive generic responses that don't match their intended outcome, causing repeated prompt refinement or switching to other AI tools. Solving this improves the core AI experience and supports long-term retention.	**High**
2	If NotebookLM improves cross-document reasoning and synthesis, users will discover deeper insights across uploaded sources.	Strengthens NotebookLM's research capabilities but addresses a narrower use case compared to personalized outputs.	**Medium-High**